SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Rakshana AR (192126007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AR.R

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

Big Data Architecture For Retail Analytics:

1) Cloud-Based Big Data Architecture:

A retail company can use cloud-based big data technology to analyze large volumes of customer transaction data volumes of customer transaction data collected from online & offline stores.

- Data Ingestion: Apache Kafka collects transaction data from different sources.
- Distributed Storage: Cloud storage services such as Amazon S3 store raw and processed data.
- Data Processing: Apache Spark process large datasets using distributed computing.
- Analytics: Cloud data warehouses and visualization tools help identify Purchasing Patterns & Improve product recommendations.

2) Integration of Batch & stream processing:

- Batch processing: Process historical sales data at scheduled intervals to identify long-term purchasing trends.
- Stream Processing: Analyzes live transactions continuously to identify current customer preferences.
- Integration: Apache Spark supports Batch Processing while Apache Kafka & Apache Flink can support real-time process.

3. Data Lakes & Data warehouses:

A data lake stores large amounts of raw data in different formats including transaction records & customer activity data.

A data where we store must be safe, so a data warehouse stores cleaned & structured data for reporting and analysis. It supports flexible analysis & accurate business reporting.

4) Scalability & cost Efficiency:

- Auto-scaling - Automatically adjusts computing resources according to demand.
- Data compression: Reduces storage requirements.
- Storage Tiering: Moves older data to lower cost storage
- Serverless processing: Reduces costs by charging according to resource usage.
- Resource Monitoring: Identifies unnecessary resource consumption.

Secure Cloud Architecture for Healthcare Data:

1) Secure Cloud Architecture:

A healthcare organization must protect patient records and medical images while allowing authorized doctors to access them securely from different locations.

The architecture should include:

Cloud storage: stores patient records & medical images securely.

Private networks: protects communication between hospitals and cloud services.

Identity & Access Managment (IAM): Ensures that only authorized users can access Patient information.

Backup & Recovery: Maintain encrypted backups to recover patient data after accidental deletion or system failure.

2) Encryption, IAM, & Key Managment Encryption:

- Use encryption at rest to protect patient records and medical images stored in the cloud.
- Use TLS encryption for data transmitted between doctors, hospitals, & cloud servers.
- Encryption helps prevent unauthorized users from reading sensitive information.

Identity & Access Management (IAM):

- Assign permissions according to each users role & responsibilities.

- Apply the principle of least privilege so users receive only the access need.
- Use multi-factor authentication (MFA) to provide additional account prote
- Review user permissions regularly & immediately revoke access when employees leave the organization.

3) Auditing, Logging & Compliance Monitoring:

- Audit Logs: Record who accessed Patient records, what actions they performed & when the access occured.
- Activity Monitoring: Monitoring login attempts, file downloads, permission changes, & unusual access patterns
- Automated Alerts: Generate alerts for suspicious activities, repeated failed logins, or unauthorized access attempts.
- Compliance Monitoring: Regular check cloud configurations and security policies against applicable healthcare privacy regulations.
- Log Protection: Restrict access to audit logs & protect them against unauthorized modification or deletion.

4) Preventing Unauthorized Access & Data Breaches:

1) Multi-factor Authentication:- Require additional verification when users sign in.
2) Least-Privilege Access :- Grant only the minimum permissions necessary for each role.
3) Network Security :- Use firewalls, Private endpoints & network segmentation to restrict acces.
4) Data loss prevention:- Monitor & control unauthorized transfer of sensitive patient information.
5) Threat Detection :- Use security monitoring tools to detect suspicious activities & Potential attacks.
6) Regular Updates :- Patch operation systems, applications & security softwares to fix vulnerabilities.
7) Backup & Recovery : Maintain encrypted backups & test recovery Procedures regularly.

Serverless Architecture for an Online Ticket Booking Platform:

1) Serveless Cloud Architecture:

A serverless architecture allows the tickets booking company to run its application without directly managing servers. The architecture includes:

- API Gateway: Receives booking requests from web & mobile applications.
- Serverless Functions: AWS Lambda or Azure Functions process ticket searches, bookings, and cancellations.
- Managed Database: stores event details, ticket availability, & customer information.
- Message Queues: Handle background tasks such as sending booking conformations.
- Monitoring Tools: Track application performance & errors.

2) Event-Driven Computing & scalability:

- Functions automatically scale according to incoming requests, subject to service limits.
- Message queues manage sudden increases in requests.
- Background tasks can run asynchronously without delaying the main booking process.
- Event triggers automatically execute tasks when specific events occur.

For example, when tickets for a popular concert become available, serverless functions can scale to handle increased booking requests.

3) Role of Managed Databases, API Gateways & Message Queues.

- Managed Database: Store booking information & ticket availability reducing database maintenance work.
- API Gateways: Provide secure access to application services & support authentication & request throttling.
- Message Queues: Manage large volumes of tasks & separate background Processing from Critical booking operations.

4) Advantages & Limitations of serverless Computing:

Advantages:

- Automatic scaling during traffic spikes
- Reduced server maintenance.
- Pay-per-use pricing
- Faster application development
- Easy integration with event-driven services.

Limitations:

- Cold starts can increase response time
- Function execution limits may affect long-running tasks.
- Costs may increase unexpectedly during heavy traffic.
- Debugging distributed applications can be difficult.
- Dependence on a specific cloud provider may cause vendor lock-in.

Cloud-Based ETL/ELT Pipeline for Enterprise Data Integration:

1) Design of a Cloud-Based ETL/ELT pipeline:

1. Extraction: Collect data from databases, applications, APIs & files.
2. Ingestion: Transfer the collected data to cloud storage.
3. Transformation: Clean, validate, standardize, & combine data from different sources.
4. Loading: Store the processed data in a cloud data warehouse. In ELT, data is loaded first and transformed within the warehouse.
5. Analytics: Use bussness intelligence tools to generate reports & dashboards.

2) Data Quality, Metadata Management, & Governance:

Data Quality:

- Remove duplicate records.
- Identify missing & incorrect values.
- Standardize dates, currencies, and formats.
- Validate data before processing.

Metadata Management:

- Maintain information about data sources, formats & ownership.
- Use a data catalog to help users find & understand datasets.
- Track data lineage to understand how data is transformed.

Data Governance:

- Define data ownership & access permissions.
- Apply data security & privacy policies.
- Establish retention policies and conduct regular audits.

3) Role of workflow Orchestration & scheduling Tools:

They help to:

- schedule data processing jobs.
- Execute tasks in the correct order.
- Manage dependencies between tasks.
- Retry failed jobs automatically.
- Monitor pipeline execution & generate alerts.

4) Query performance & storage cost optimization:

- Partitioning: Divides large datasets by date or region to improve query performance.
- Compression: Reduces storage requirements.
- Columnar Formats: Formats such as Parquet improve analytical Processing efficiency.
- Query Optimization: Avoids unnecessary columns & Processes only required data.
- Storage Tiering: Moves infrequently accessed data to lower-cost storage.
- Resource Scaling: Adjusts computing resources according to workload.

Cloud-Based Machine Learning Architecture for Predicting Delivery Delays:

1) Cloud-Based Machine Learning & Analytics Architecture:

1. Data Ingestion: Collect shipment records, weather updates & live GPS.
2. Cloud Storage: Store raw & processed data in a cloud data lake.
3. Data Processing: Use Apache Spake to process large datasets.
4. Feature Engineering: Generate features such as distance remaining, weather conditions & historical delivery times.
5. Model Training: Train machine learning models using historical data.
6. Model Deployment: Deploy the trained models as a cloud-based Prediction service.
7. Real-Time Prediction: Analyze live data to estimate delivery times & identify possible delays.
8. Monitoring: Track model accuracy & prediction performance.

2) Role of Big Data Technologies in Training & Inference:

Model Training:

- Apache spark processes large volumes of historical shipment data.
- Distributed computing speeds up data preparation & model training.
- Historical GPS, weather, and shipment records help identify patterns associated with delivery delays.

Model Inference:

- Stream processing tools analyze live GPS & weather updates.
- The deployed model uses updated data to predict possible delays.
- Predictions are shared with logistics managers to support timely decisions.

3) Cloud-Native AI/ML services for model Deployment & Monitoring:

- Model Training: Provide managed computing resources for training.
- Model Deployment: Host trained models for real-time predictions.
- Auto-scaling: Adjust computing resources according to demand.
- Model Monitoring: Track prediction accuracy & changes in incoming data.
- Version control: Maintain model versions & allow rollback when required.
- Automated Retraining: Retrain models when new data becomes available or performance decreases.

4) Data Ingestion, Feature Engineering & Model Updates:

Data Ingestion:

- Use Apache Kafka or similar Streaming services to collect live GPS data.
- Obtain weather information through APIs.
- Validate incoming data & handle missing or duplicate records.
- Store historical & incoming data in scalable cloud storage.

Feature Engineering:

- Calculate distance remaining & estimated travel time.
- Combine weather conditions with shipment locations.
- Analyze historical delivery patterns.
- Ensure consistent & valid model inputs.